Do System One Models Make Better Rerankers?

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Abstract. System One models, or decision models, answer a question with a probability rather than generated text and have recently been adopted as rerankers. We ask whether the decision fine-tuning behind them actually improves reranking effectiveness or probability calibration over conventional reranker fine-tuning. Across six Qwen backbones of different sizes and ten TREC DL and BEIR collections, we compare zero-shot base models, decision fine-tunes, and reranker fine-tunes of the same backbone. We show that reranker fine-tuning is more effective on all five backbones for which both variants exist, including on collections outside the fine-tuning data, while for smaller backbones decision fine-tuning provides little benefit even compared to base models. Decision models are also not inherently better calibrated: their calibration depends strongly on how well the relevance question matches the evaluation labels. Overall, decision fine-tuning provides no clear advantage for reranking over dedicated reranker fine-tuning.[1]

Keywords: Reranking · Calibration · System One Models · Fine-Tuning

1 Introduction

Cross-encoders are a well-established approach to reranking first-stage retrieval candidates [16], while newer methods leverage open large language models fine-tuned for ranking tasks [14,33,34]. A recent alternative is *System One models*, or *decision models*, which answer questions about a given input with a probability for each possible answer rather than generating text [29]. Jev (TypeSafe AI, September 2026), the first such model, is already reported to be a competitive reranker [11, 23] and is evaluated or integrated in retrieval systems [9, 12, 30]. Open decision models such as Kev [19] and Cloudflare's Clef [2] followed, fine-tuned from Qwen backbones that are also used by dedicated rerankers such as Qwen3-Reranker and Prism.

One of the primary purported benefits of decision models is their ability to produce calibrated probabilities. Reranking, however, depends on the order of the scores and not on their absolute values: a monotone transformation of the probabilities leaves the ranking unchanged, so calibration alone does not make a better reranker. It is therefore unclear: how much of the reported effectiveness is due to *decision fine-tuning* of System One models and how much to the capability of the underlying language model? We argue that a fair evaluation of

[1] Repository: https://anonymous.4open.science/r/ecir2027-decision-rerank-D7C6/

decision models as rerankers requires controlling for the underlying base model and comparing against reranker fine-tunes of the same backbone. In addition, it should test whether the claimed advantage in calibration actually holds for relevance estimation. Concretely, we ask the following questions:

RQ1 Do decision models rerank better than their base models or reranker fine-tunes of the same base model?

RQ2 Are their relevance probabilities better calibrated than either?

We make two contributions. First, we provide a broad, controlled evaluation of six Qwen backbones from 0.6B to 27B parameters with their respective decision fine-tunes and reranker fine-tunes on TREC DL 2019–2023 and five BEIR collections. We show that reranker fine-tuning is more effective on all five base models where both fine-tunes exist, also on collections outside the training data of any fine-tune. Second, we show that decision models are better calibrated than their base models, but better than reranker fine-tunes of the same backbone only if the question matches the relevance labels. Their calibration thus depends on the wording, and our own reranker fine-tune is as well calibrated as the best decision model of its size.

2 Related Work

LLM-based rerankers. Neural rerankers are typically trained to score query–passage pairs using relevance supervision, first with encoder or sequence-to-sequence models [16, 17] and more recently with open LLMs [14, 33, 34]. Other approaches rerank candidates jointly, either by generating a ranked list [21, 27] or by allowing interaction between passages, as in the Set-Encoder [24]. LLMs can also be used without reranking-specific fine-tuning, for example through relevance prompting [22, 35], although their effectiveness can be sensitive to prompt formulation [26].

Decision models in retrieval. Existing evidence on decision models for retrieval is limited and largely non-peer-reviewed [9, 11, 12, 23, 30]. The open Laya model has also been used as a relevance gate [3, 25]. Prior work evaluates Jev against existing rerankers, but does not compare decision and reranker fine-tuning on the same backbone.

Calibration of relevance scores. Common ranking losses are invariant to shifts of scores, so rankers trained with them can order passages well without being calibrated [20, 31]; scores can be calibrated during or after training [10, 32].

3 Decision Models as Rerankers

A decision model takes a structured input, the *state*, together with one or more questions about it, and returns a probability for each possible answer. We use binary questions only, so that every question yields one number, the probability of "yes". Reranking follows directly: for a query q with candidates $d_1, \ldots, d_k$

from a first-stage retriever, we ask about each passage d_i whether it is relevant to q and order the passages by the returned probabilities p_i. For the state, we consider two modes. In *isolated* mode, the state contains the query and a single passage, so that $p_i = P(\text{yes} \mid q, d_i)$. This is akin to a pointwise reranker, like a cross-encoder. In *joint* mode, the state contains the query and all k candidates in first-stage order, and the k questions are asked about this one state, so that $p_i = P(\text{yes} \mid q, d_1, \ldots, d_k)$ can depend on the other candidates. Joint mode shares its input with listwise rerankers but returns a score per passage instead of a permutation, like the Set-Encoder [24].

4 Experimental Setup

Rerankers. We study six Qwen backbones: Qwen3-0.6B and 8B, Qwen3.5-0.8B, 4B and 9B, and Qwen3.8-27B. Each is evaluated in up to three variants. The *zero-shot backbone* (base model in the tables) is the released model, asked the question in its chat format with thinking off and scored by the probability of "yes" as the first answer token. The *decision fine-tunes* are Kev [18,19] (LoRA up to 9B, a full fine-tune at 27B), Clef-Flash (9B) and Clef (27B) [2], each run in isolated and joint mode (Sect. 3). The *reranker fine-tunes* are Qwen3-Reranker [34], built on the same pretrained checkpoints as Kev, and Prism [33], both scored by the probability of "yes" against "no"; No reranker fine-tune of Qwen3.8-27B has been released, so this variant is missing at 27B. As a control, we also train a reranker ourselves (*Ours-9B*): Qwen3.5-9B fine-tuned with LoRA on MS MARCO only, with a ranking loss and a yes/no classification loss. This gives us one reranker fine-tune whose training data and losses we control. Qwen3-Reranker, Prism and Ours-9B saw MS MARCO in training; Kev's training data contain neither MS MARCO nor BEIR [18]. As reference outside the controlled comparison, we report Jev 1.13.0 and BM25. The results of other established rerankers are reported in our public repository.

Questions. Decision fine-tunes and base models are asked S1Rank's question [23]: whether the passage is relevant to the query, i.e., answers it or contains information that directly addresses it. For calibration, we also ask a stricter question, whether the passage answers the query, which matches relevance from grade 2. Both were fixed before running the experiments.

Collections and candidates. We use TREC DL 2019–2023 [4–8] on the MS MARCO v1 and v2 passages [15] and five BEIR collections [28]: TREC-COVID, NFCorpus, SciFact, Touché and DBPedia. None of the fine-tunes was trained on TREC-COVID, Touché or DBPedia, which form the *zero-shot subset* (ZS). Every system reranks the BM25 top-100 of Pyserini [13]; these are the same candidates used by S1Rank [23] and in earlier comparisons [1,21,27].

Evaluation. Ranking effectiveness is measured by nDCG@10, with paired t-tests over queries, Holm-corrected across the ten collections ($p < 0.05$). Calibration is measured by the expected calibration error (ECE, 15 bins) [10] and the Brier score, i.e. the mean squared error of the probability against the binary label. Both are computed on the judged query–passage pairs of TREC DL

Table 1. Base model, decision fine-tune and reranker fine-tune of the same foundation model (nDCG@10, ×100, BM25 top-100). **Avg:** mean over all ten collections, including DL20 and DL22, which are not shown for space; **ZS:** mean over the collections without a training split (TREC-COVID, Touché, DBPedia). Per foundation model and column: best in **bold**, second best underlined. Training: ○ base model, ▲/△ decision fine-tune (joint/isolated), ■ reranker fine-tune (rows shaded grey, red, blue); Paired t-test, Holm-corrected over the ten collections, $p < 0.05$: * reranker fine-tune vs. the decision fine-tune with the best Avg, † that decision fine-tune vs. the base model, ‡ joint vs. isolated. No reranker fine-tune of the 27B model exists. Jev: closed model, shown for reference.

			TREC DL			BEIR					Mean	
	System	Mode	DL19	DL21	DL23	COVID	NFC	SciF	Touché	DBP	Avg	ZS
Qwen3-0.6B												
○	Qwen3-0.6B		32.6	32.5	14.3	50.3	16.2	18.8	11.5	14.3	23.8	25.4
▲	Kev-0.6B	joint	18.4‡	19.1‡	11.0‡	34.0‡	12.0‡	1.1‡	4.5‡	6.1‡	13.2	14.9
△	Kev-0.6B	isolated	33.4	39.8	18.8	58.2	17.0	16.2	13.0	21.5†	27.0	30.9
■	Qwen3-Reranker-0.6B		**69.5***	**65.2***	**38.8***	**86.2***	**36.4***	**76.6***	**35.7***	**42.7***	**56.1**	**54.9**
Qwen3-8B												
○	Qwen3-8B		70.4	65.8	42.1	80.2	34.1	72.8	24.0	37.1	53.3	47.1
▲	Kev-8B	joint	63.4‡	60.1	35.7	48.2‡	19.8‡	19.2‡	9.7‡	36.7‡	38.8	31.5
△	Kev-8B	isolated	71.2	61.4†	37.6†	80.8	35.2	73.9	26.5	40.0†	52.7	49.1
■	Qwen3-Reranker-8B		**73.2**	**67.0***	**45.2***	**84.0**	**37.3***	**79.9***	**43.1***	**46.2***	**59.4**	**57.8**
Qwen3.5-0.8B												
○	Qwen3.5-0.8B		47.7	53.3	27.9	65.3	20.4	41.1	21.9	25.8	37.6	37.7
▲	Kev-0.8B	joint	35.7	38.6‡	18.2‡	45.3‡	18.6	20.2‡	9.4‡	14.6‡	25.3	23.1
△	Kev-0.8B	isolated	42.8	48.3	23.8	71.9	20.9	36.0	18.0	32.2†	36.0	40.7
■	Prism-0.8B		**72.3***	**66.3***	**43.7***	**84.2***	**36.1***	**76.3***	**35.6***	**41.7***	**57.1**	**53.8**
Qwen3.5-4B												
○	Qwen3.5-4B		69.0	66.2	40.1	79.5	34.1	72.9	24.8	37.0	52.7	47.1
▲	Kev-4B	joint	65.5	62.4	38.3	75.3‡	29.3‡	62.4‡	23.7	40.7	49.8	46.6
△	Kev-4B	isolated	62.6†	62.0	35.4	82.6	34.1	70.7	23.0	39.7†	51.3	48.4
■	Prism-4B		**74.4***	**70.3***	**46.2***	**84.6**	**37.4***	**76.9***	**34.3***	**43.7***	**58.7**	**54.2**
Qwen3.5-9B												
○	Qwen3.5-9B		69.6	67.7	38.1	81.2	33.6	75.0	27.5	35.8	53.7	48.2
▲	Kev-9B	joint	63.1	63.6	39.1	78.2	31.5	60.9‡	22.6	38.5‡	50.2	46.4
△	Kev-9B	isolated	65.7	60.8	35.4	77.7	32.8	72.6	19.8	34.1	50.1	43.8
▲	Clef-Flash	joint	68.8	64.3	40.4	82.1	35.1‡	65.4	21.3	38.8‡	52.8	47.4
△	Clef-Flash	isolated	68.8	66.4	43.8†	82.3	**37.3**†	68.7†	22.1	42.6†	54.7	49.0
■	Prism-9B		**74.0***	**71.0***	44.2	83.0	37.2	**79.8***	34.7*	43.2	59.0	53.7
■	*Ours-9B*		73.8*	67.3	**47.0**	**84.2**	37.1	77.9*	**36.5***	**47.7***	**59.4**	**56.1**
Qwen3.8-27B												
○	Qwen3.8-27B		71.6	68.5	44.1	83.0	37.6	78.5	30.2	45.0	57.4	52.7
▲	Kev-27B	joint	**73.9**	69.9	**49.3**†	**85.9**†	**38.7**†	**81.1**†	**40.8**‡	46.9†	**61.1**	**57.9**
△	Kev-27B	isolated	73.4	**70.1**	46.8	85.5	38.3	79.9	34.0	**47.5**	59.8	55.7
▲	Clef	joint	71.4	68.6	45.9	82.9	36.3	62.9‡	30.8	42.8	56.1	52.2
△	Clef	isolated	69.4	66.7	42.9	83.5	37.8	71.9	33.9	44.1	56.4	53.8
References												
	BM25		50.6	44.6	26.2	59.5	32.2	67.9	**44.2**	31.8	43.2	45.2
▲	Jev	joint	**73.3**	**72.2**	**46.8**	**85.0**	38.2	**81.0**	39.9	**46.8**‡	**60.5**	**57.2**
△	Jev	isolated	72.6	70.3	44.7	**85.0**	**38.3**	79.7	40.3	44.8	59.5	56.7

2019–2023, the only collections that also judge non-relevant passages. A passage counts as relevant from grade 2. We also test one use case of the probabilities, the *Count* error: the sum of a query's probabilities over its judged passages estimates how many of them are relevant; we report the mean absolute error of this estimate, in passages.

5 RQ1: Decision vs. Reranker Fine-Tuning

Reranker fine-tuning wins on every base model. On all five backbones for which both exist, the reranker fine-tune is more effective than the decision fine-tune in either mode. Averaged over the ten collections, its margin over the better decision mode ranges from 4.3 points nDCG@10 at Qwen3.5-9B to 29.1 at Qwen3-0.6B. The difference is significant on 46 of the 60 pairs of model and collection, and the decision fine-tune is never significantly better (*). The margin is not an artefact of evaluating on MS MARCO: on the zero-shot subset, which no fine-tune was trained on, it ranges from 4.7 to 24.0 points. Nor does it depend on the specialized training recipe of the published rerankers, as Ours-9B, trained on MS MARCO alone with standard losses, matches Prism-9B on average (59.4 vs. 59.0) and exceeds it on the zero-shot subset (56.1 vs. 53.7). Kev-27B is only 1.7 points ahead of reranker fine-tunes a third of its size, and a gain as on the smaller backbones would place a 27B reranker fine-tune ahead of it.

Decision fine-tuning barely improves or even degrades on its base model. Up to 9B, the better mode of a decision fine-tune stays within 3.5 points of its zero-shot backbone, with significant differences on at most half of the collections (†), whereas reranker fine-tuning adds at least 5 points. The 27B backbone is the exception: Kev-27B improves on it significantly on most collections. Clef, the other 27B decision fine-tune, however, does not improve.

Joint scoring helps only the largest models. Scoring all candidates in one state gains about one point for Kev-27B and Jev (Table 1) but costs Kev-0.6B, Kev-0.8B and Kev-8B more than ten (‡). A likely reason is state length: a joint state has 13k tokens or more, whereas Kev-0.8B and Kev-4B were trained on states of at most 7.5k tokens and Kev-27B on up to 32k [18].

6 RQ2: Are Decision Models Better Calibrated?

Both types of fine-tuning improve calibration over the base model. As expected, the zero-shot base models are poorly calibrated, especially the smallest ones (ECE up to 0.750; Table 2). While decision fine-tuning lowers ECE and Brier score on every backbone, in line with the claim made for decision models, so does reranker fine-tuning.

With S1Rank's question, decision fine-tunes are not better calibrated than reranker fine-tunes. On every backbone with both fine-tunes, a reranker fine-tune has the lower Brier score, and its ECE is lower or close. The one clear advantage of a decision fine-tune, Clef-Flash over Prism-9B, disappears against Ours-9B (ECE 0.111 vs. 0.140).

Calibration depends on matching the question to the labels. With the stricter question (Answers? in Table 2), the ECE of almost every decision and base model drops, with hardly any change in ranking. In this setting, the decision fine-tunes are better calibrated than the published reranker fine-tunes on the 4B, 8B and 9B backbones. The best calibrated systems in the study are Jev and the two 27B decision fine-tunes. Two observations qualify this. First, the base models

Table 2. Calibration on the judged pairs of TREC DL 2019–2023 (means; lower is better; Count: error in passages per query, always guessing the median count: 8.6). Rows name the variant of the foundation model in the group header; decision fine-tunes in their better mode (j/i). *Relevant?*: S1Rank's question; *Answers?*: does the passage answer the query (rerankers ask no question: their values repeated in grey).

	Relevant?			Answers?	
System	ECE	Brier	Count	ECE	Count
Qwen3-0.6B					
∘ base model	.750	.750	36.5	.750	36.5
△ Kev (i)	.579	.543	28.2	**.503**	**24.7**
■ Qwen3-Reranker	**.511**	**.481**	**24.8**	.511	24.8
Qwen3-8B					
∘ base model	.383	.380	17.5	.231	8.7
▲ Kev (j)	**.243**	.248	**12.8**	**.113**	**7.6**
■ Qwen3-Reranker	.257	**.238**	12.9	.257	12.9
Qwen3.5-0.8B					
∘ base model	.716	.694	34.8	.617	29.9
△ Kev (i)	.357	.309	18.0	.336	16.9
■ Prism	**.243**	**.193**	**12.7**	.243	12.7
Other systems					
▲ Jev (j)	**.224**	**.187**	**11.3**	**.075**	**6.3**

	Relevant?			Answers?	
System	ECE	Brier	Count	ECE	Count
Qwen3.5-4B					
∘ base model	.296	.269	14.7	.181	9.8
△ Kev (i)	**.216**	.207	**11.5**	**.132**	**8.4**
■ Prism	.218	**.172**	**11.5**	.218	11.5
Qwen3.5-9B					
∘ base model	.283	.265	14.1	.165	8.7
▲ Kev (j)	.237	.209	12.2	.160	9.5
▲ Clef-Flash (j)	.140	.165	8.9	.112	8.1
■ Prism	.242	.183	12.3	.242	12.3
■ *Ours*	**.111**	**.157**	**7.3**	.111	**7.3**
Qwen3.8-27B					
∘ base model	.166	.180	8.9	.117	**5.8**
▲ Kev (j)	.156	.159	8.5	**.076**	6.0
▲ Clef (j)	**.147**	**.157**	**8.4**	.091	6.7

improve as well (Qwen3-8B from 0.383 to 0.231), so part of the gain comes from the question, not from decision fine-tuning. Second, our reranker fine-tune, which asks no question, is as well calibrated as the best decision fine-tune of its size (Ours-9B 0.111 vs. Clef-Flash 0.112).

Calibration buys little in practice. When counting a query's relevant passages with S1Rank's question, no decision model is clearly better than always guessing the median count (Clef 8.4 vs. 8.6). With the stricter question, most decision models with 4B parameters or more beat this guess. Decision fine-tuning is not what makes counting work, however: the untuned Qwen3.8-27B counts most accurately of all (5.8), and our reranker fine-tune counts better than the decision fine-tune on the same backbone.

7 Conclusion

Up to 9B parameters, decision fine-tuning buys almost nothing over simply asking the base model. Its effectiveness stays within 3.5 points nDCG@10 of the base model, while reranker fine-tuning of the same backbones adds at least 5 points. Calibration, the main promise of decision models, is largely a property of the question rather than of decision fine-tuning. When the question matches the relevance labels, even base models become better calibrated, and our reranker fine-tune, which asks no question, is as well calibrated as the best decision model of its size. *Limitations:* We measured calibration only on TREC DL and at a single relevance threshold. The compared fine-tunes differ in training data, so our results describe the released models rather than decision fine-tuning as such. At 27B, where Kev is the best system, no reranker fine-tune exists for comparison.

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